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Part A — User and System Investigation

- Determine the username of the account you are currently using.
whoami
- Display your UID, primary GID, and supplementary groups.
id
- Display all users currently logged in to the system.
who
- Determine what activities the currently logged-in users are performing.
w
- Examine /etc/passwd and identify the following information for your current account:
Username
UID
GID
Home directory
Login shell

wc -l /etc/passwd
- Determine how many user-account entries are present in /etc/passwd.
wc -l /etc/passwd
- Display only the username and UID fields from /etc/passwd.
cut -d: -f1,3 /etc/passwd
- Find all accounts whose UID is 0. Explain the security significance of UID 0.
awk -F: '\$3 == 0 {print \$1, \$3}' /etc/passwd
O/P: root 0

Security significance: UID 0 represents the root/superuser account. So unauthorized accounts with UID 0 are major security risk.

A UID 0 has very high privileges that are:

- Read or modify protected files
- Change system configurations
- Create/delete users
- Start/stop system services
- Access resources restricted to normal users

➤ Compare `/etc/passwd` and `/etc/shadow`. Why are password hashes stored in `/etc/shadow` instead of `/etc/passwd`?

`/etc/passwd`

- Contains usernames, uid,gid, home directories and shells
- Readable by normal users

`/etc/shadow`

- Contains password hash and password-aging information
- Readable only by root/authorized processes

`/etc/passwd` needs to be readable by many system programs because it contains basic account information. If password hashes are stored there, any local user who can read the file can get the hashes and attempt offline password cracking. Moving password hashes to `/etc/shadow` gives them restricted permissions, reducing the risk of unauthorized access.

```
saurov@DESKTOP-VBG03KL:~$ whoami
saurov
saurov@DESKTOP-VBG03KL:~$ id
uid=1000(saurav) gid=1000(saurav) groups=1000(saurav),4(adm),24(cdrom),27(sudo),
saurov@DESKTOP-VBG03KL:~$ who
saurov pts/1 2026-09-01 13:27
saurov@DESKTOP-VBG03KL:~$ w
 13:35:21 up 7 min, 1 user, load average: 0.00, 0.01, 0.00
USER      TTY      FROM          LOGIN@      IDLE        JCPU       PCPU       WHAT
saurov    pts/1    -             13:27       7:57        0.04s      0.03s      -bash
saurov@DESKTOP-VBG03KL:~$ grep "^$(whoami):" /etc/passwd
saurov:x:1000:1000:,,,:/home/saurav:/bin/bash
saurov@DESKTOP-VBG03KL:~$ wc -l /etc/passwd
28 /etc/passwd
saurov@DESKTOP-VBG03KL:~$ cut -d: -f1,3 /etc/passwd
root:0
daemon:1
bin:2
sys:3
sync:4
games:5
man:6
lp:7
mail:8
news:9
uucp:10
proxy:13
www-data:33
backup:34
list:38
irc:39
_apt:42
nobody:65534
systemd-network:998
systemd-timesync:996
dhcpcd:100
messagebus:101
syslog:102
systemd-resolve:991
uidd:103
```

Part B — Creating and Verifying Users

- Create the following users and ensure that a home directory is created for each:alice, bob, Charlie
`sudo useradd -m alice`
- Verify that all three accounts have been added to `/etc/passwd`.
`grep -E '^(alice|bob|charlie):' /etc/passwd`
- Determine the UID assigned to each newly created user.
`id alice`
- Determine the home directory and default shell assigned to alice.
`getent passwd alice | cut -d: -f6,7`
- Inspect `/home` and verify that the required home directories were created.
`ls -l /home`
- Set passwords for all three accounts.
`sudo passwd charlie`
- Switch from your current account to alice using a login shell. Verify the change using an appropriate command, and then return to your original account.
`su - alice`
`whoami`
`pwd`
`exit`

```
saaurav@DESKTOP-VBG03KL:~$ grep -E '(alice|bob|charlie):' /etc/passwd
saaurav@DESKTOP-VBG03KL:~$ sudo useradd -m alice
saaurav@DESKTOP-VBG03KL:~$ sudo useradd -m bob
saaurav@DESKTOP-VBG03KL:~$ sudo useradd -m charlie
saaurav@DESKTOP-VBG03KL:~$ grep -E '(alice|bob|charlie):' /etc/passwd
alice:x:1001:1002::/home/alice:/bin/sh
bob:x:1002:1003::/home/bob:/bin/sh
charlie:x:1003:1004::/home/charlie:/bin/sh
saaurav@DESKTOP-VBG03KL:~$ id alice
uid=1001(alice) gid=1002(alice) groups=1002(alice)
saaurav@DESKTOP-VBG03KL:~$ id bob
uid=1002(bob) gid=1003(bob) groups=1003(bob)
saaurav@DESKTOP-VBG03KL:~$ id charlie
uid=1003(charlie) gid=1004(charlie) groups=1004(charlie)
saaurav@DESKTOP-VBG03KL:~$ getent passwd alice | cut -d: -f6,7
/home/alice:/bin/sh
```

```
saaurav@DESKTOP-VBG03KL:~$ ls /home
alice bob charlie saaurav
saaurav@DESKTOP-VBG03KL:~$ sudo passwd alice
New password:
Retype new password:
passwd: password updated successfully
saaurav@DESKTOP-VBG03KL:~$ sudo passwd bob
New password:
Retype new password:
passwd: password updated successfully
saaurav@DESKTOP-VBG03KL:~$ sudo passwd charlie
New password:
Retype new password:
passwd: password updated successfully
saaurav@DESKTOP-VBG03KL:~$ su - alice
Password:
su: Authentication failure
saaurav@DESKTOP-VBG03KL:~$ su - alice
```

Part C — Group Management and Access Control

- Create two groups: security and developers
`sudo groupadd security`
- Add alice and charlie to the security group.
`sudo usermod -aG security alice`
- Add bob to the developers group.
`sudo usermod -aG developers bob`
- Without switching accounts, verify the group membership of all three users.
`groups alice`
`groups bob`
- Display the entries corresponding to security and developers from `/etc/group`.
`grep -E '^(security|developers):' /etc/group`
- Add bob to the security group without removing him from the developers group.
`sudo usermod -aG developers bob`
- Verify that bob now belongs to both groups.

```
saaurav@DESKTOP-VBG03KL:~$ sudo groupadd security
saaurav@DESKTOP-VBG03KL:~$ sudo groupadd developers
saaurav@DESKTOP-VBG03KL:~$ getent group security
security:x:1005:
saaurav@DESKTOP-VBG03KL:~$ sudo usermod -aG security alice
saaurav@DESKTOP-VBG03KL:~$ sudo usermod -aG security charlie
saaurav@DESKTOP-VBG03KL:~$ sudo usermod -aG developers bob
saaurav@DESKTOP-VBG03KL:~$ groups alice
alice : alice security
saaurav@DESKTOP-VBG03KL:~$ groups bob
bob : bob developers
saaurav@DESKTOP-VBG03KL:~$ groups charlie
charlie : charlie security
saaurav@DESKTOP-VBG03KL:~$ grep -E '^(security|developers):' /etc/group
security:x:1005:alice,charlie
developers:x:1006:bob
saaurav@DESKTOP-VBG03KL:~$ █
```